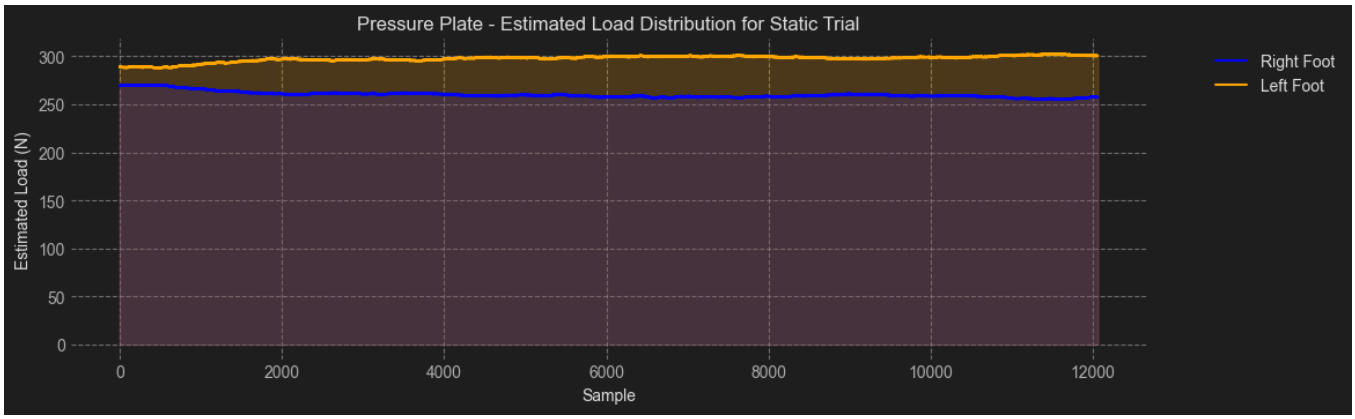
XSensor vs Pressure Plate

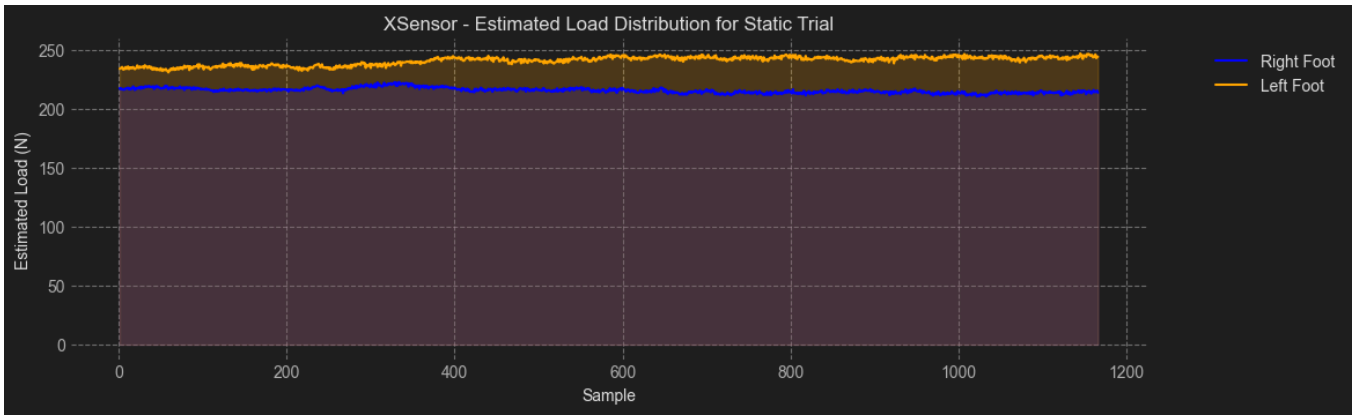
Comparing load distribution and force readings between the AMTI pressure plate and XSensor insoles across three trials: static standing, weight shifting, and walking.

Raw Data - Individual Device Readings

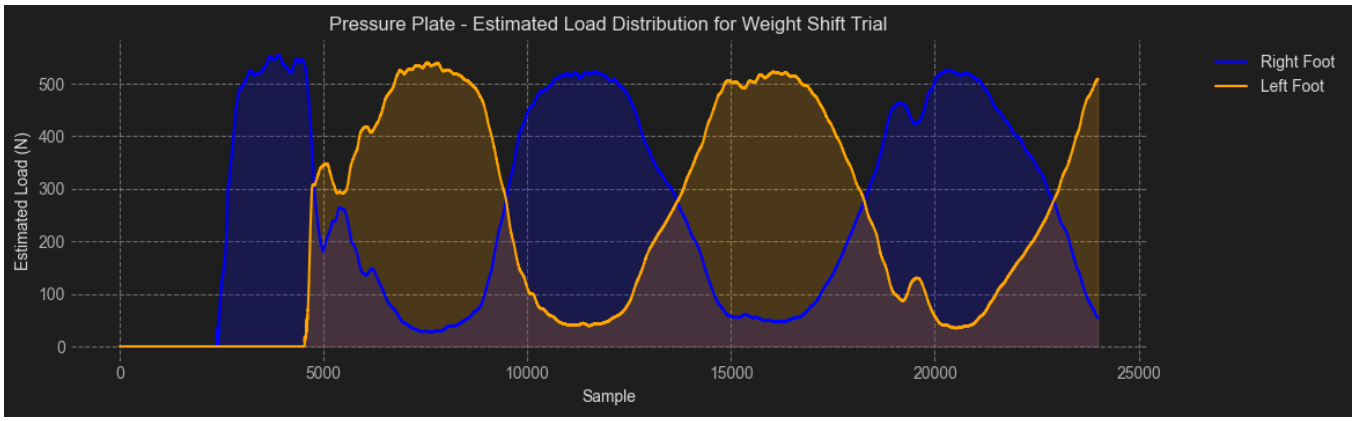
Each trial plotted separately for the pressure plate and XSensor, showing left and right foot force over time.

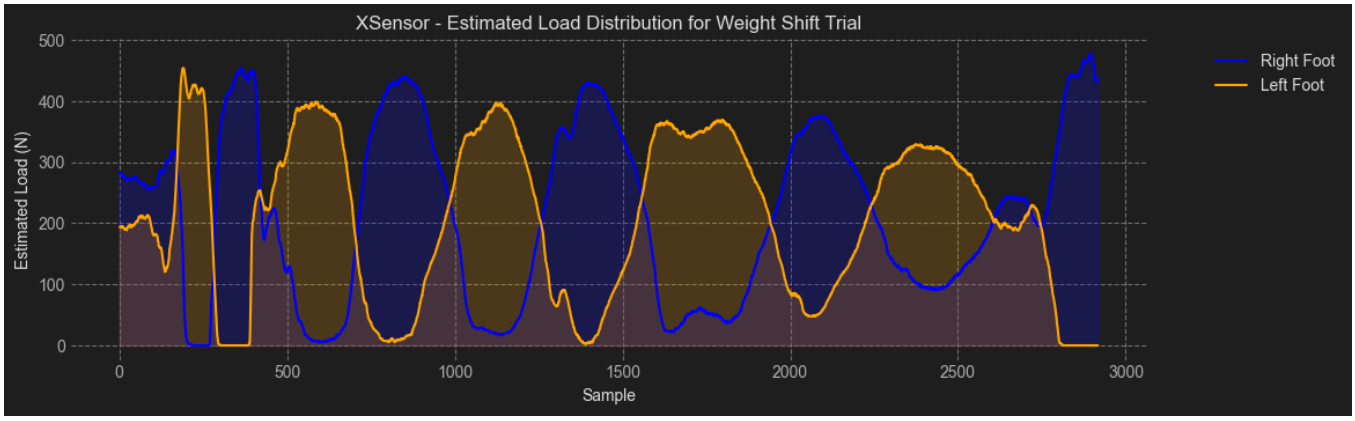
Static Trial





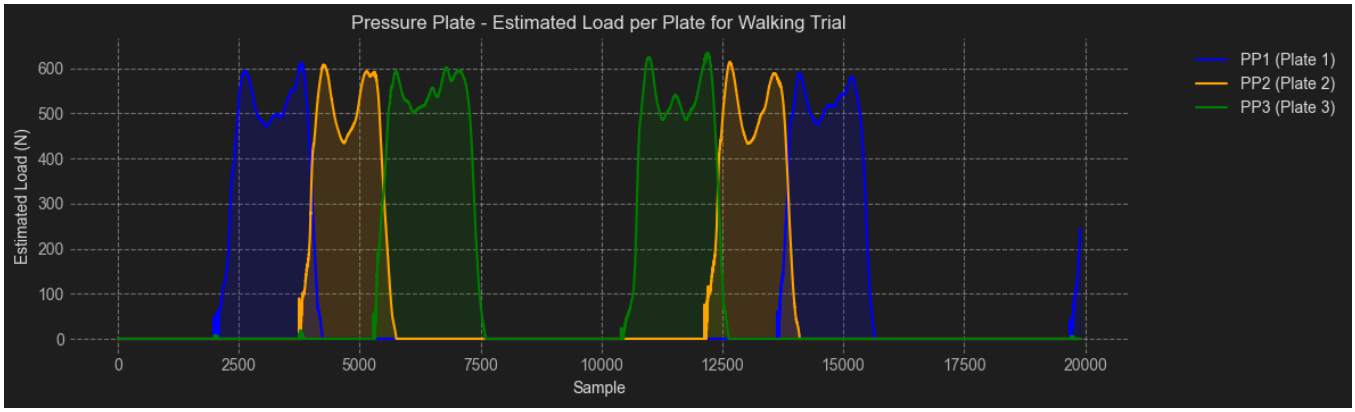
Weight Shift Trial

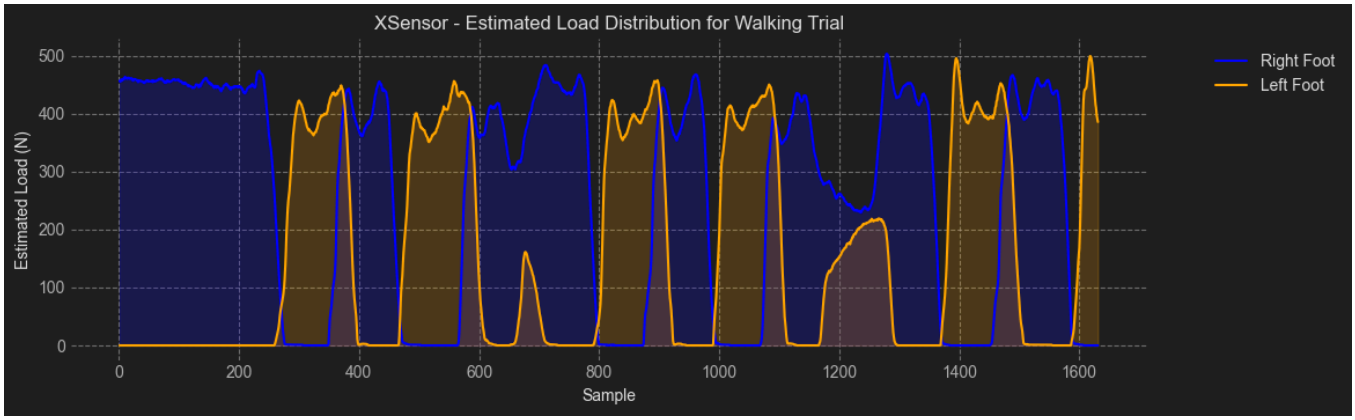




Walking Trial

The walking trial uses 3 pressure plates. The subject walked forward stepping PP1 → PP2 → PP3, turned around, then stepped PP2 → PP1. This means **PP1 and PP3 always recorded the same foot**, and **PP2 recorded the other foot**. It is unknown which foot (left or right) started on PP1, but for comparison with XSensor we assume **PP1 & PP3 = Left Foot** and **PP2 = Right Foot**.





Data Processing & Comparison

To compare the two devices fairly, we need to handle three things:

1. Match the sampling rates - The pressure plate records at 2400 Hz and the XSensor at 150 Hz. That's 16x more data points from the PP. We downsample the PP by averaging every 16 samples into 1, bringing it down to 150 Hz.

2. Align the recordings in time - Two different people pressed "start" on each device, so there's a timing offset. We use cross-correlation to find the best alignment (see below).

3. Trim to the overlapping window - After aligning, we cut both signals to only the time period where both were actively recording.

Finding the Best Alignment

This is achieved using cross-correlation. It tries every possible offset - shifting one signal 1 sample, 2 samples, 3 samples, etc. At each position it calculates a score for how well the two signals match. The offset with the highest score is the lag. This is implemented by:

1. Normalizing both signals (subtract mean, divide by standard deviation) so we're comparing shape, not magnitude.
2. Sliding one signal across the other - `np.correlate(a, b, mode='full')` computes the dot product at every possible offset. When the signals line up well, the dot product is large.
3. Finding the peak - `np.argmax(correlation)` gives the offset where the two signals overlap best. We subtract `len(b) - 1` to convert from the array index to an actual lag value (positive = PP started earlier, negative = XS started earlier).

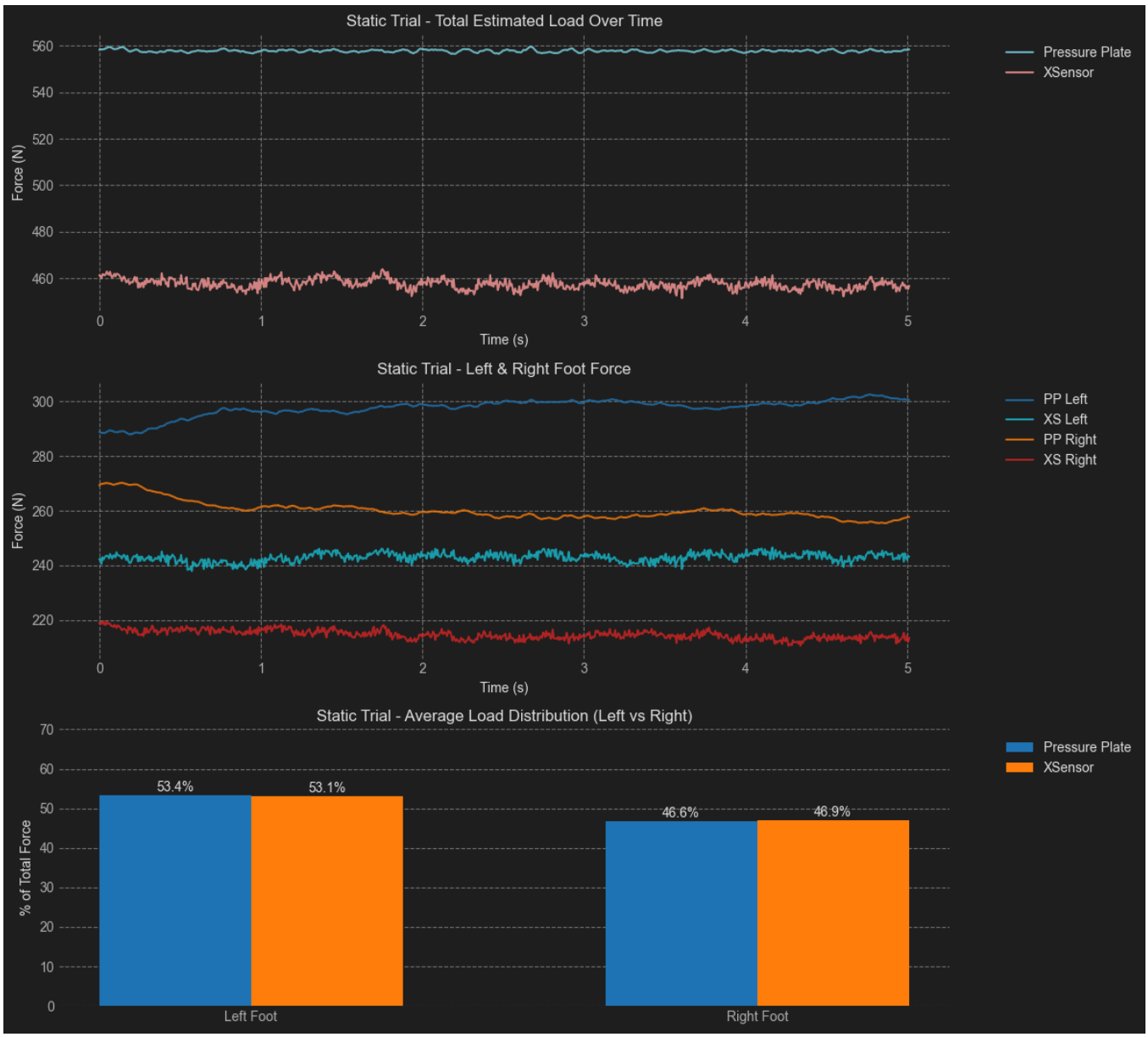
Aligning and Trimming

Once we know the lag (time offset in # of samples, eg. lag of 206 samples at 150 Hz = 1.37s time offset) from the function that finds best alignment:

1. Shift the earlier signal - If lag is positive (PP started first), we skip ahead that many samples in the PP data. If negative (XS started first), we skip ahead in the XS data.
2. Trim to equal length - After shifting, one signal may still be longer than the other. We cut both to the length of the shorter one so every time point has data from both devices.
3. Add a time column - Convert sample indices to seconds (dividing by 150 Hz) for plotting.

Static Trial - Aligned Comparison

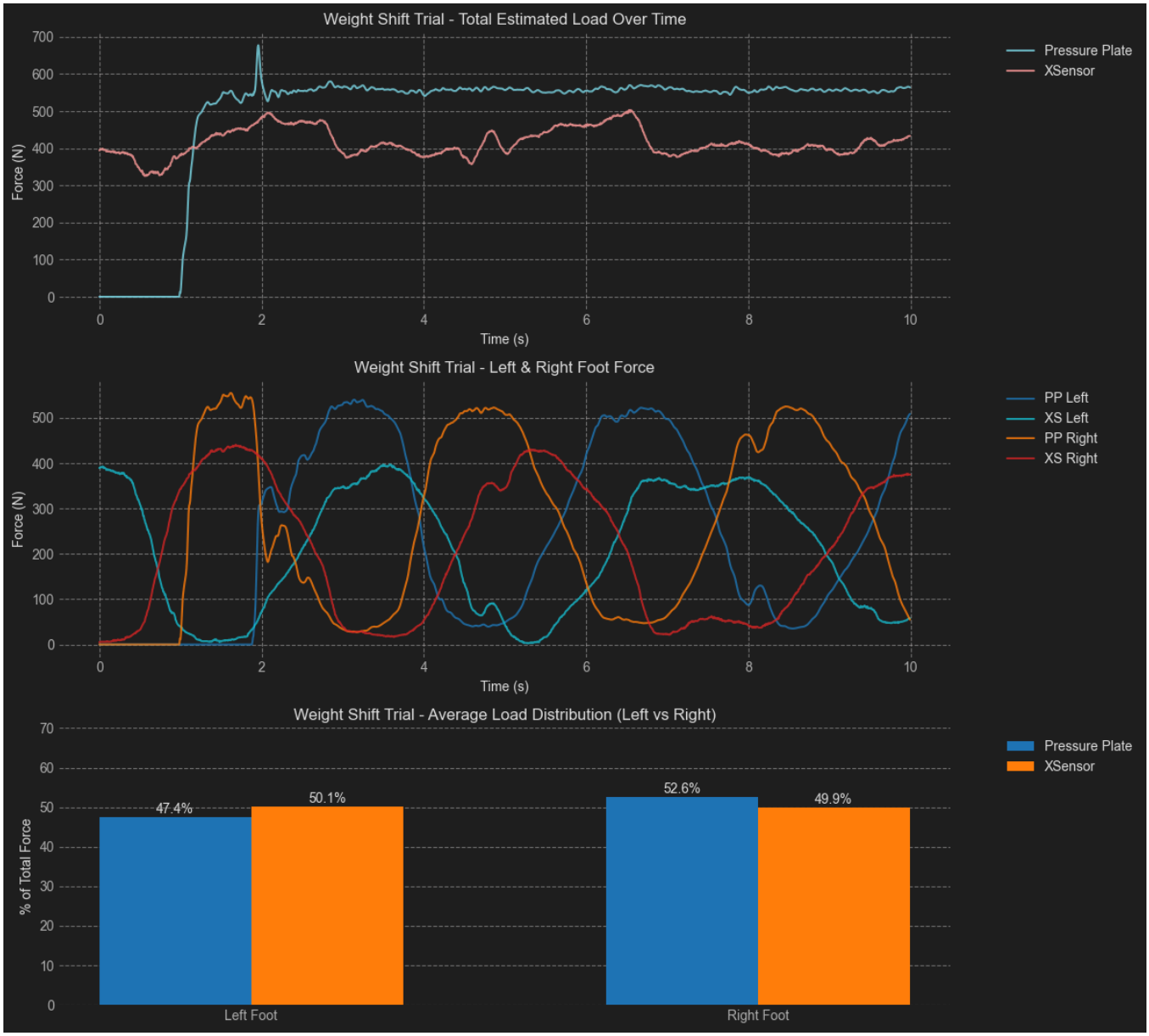
```
Alignment: XS started earlier by 2.54s (381 samples)
Comparing 5.0s of overlapping data
```




```
Avg Total Force  - PP: 557.7 N, XS: 457.6 N
XS reads 100.1 N lower (18.0%)
Left/Right Split - PP: 53.4% / 46.6%
                   XS: 53.1% / 46.9%
```

Weight Shift Trial - Aligned Comparison

```
Alignment: XS started earlier by 3.98s (597 samples)
Comparing 10.0s of overlapping data
```




```
Avg Total Force  - PP: 494.2 N, XS: 415.3 N
XS reads 78.9 N lower (16.0%)
Left/Right Split - PP: 47.4% / 52.6%
                   XS: 50.1% / 49.9%
```

Walking Trial - Aligned Comparison

```
Alignment: XS started earlier by 1.19s (178 samples)
Comparing 8.3s of overlapping data
```

```
Avg Total Force  - PP: 276.5 N, XS: 421.1 N
XS reads 144.6 N higher (52.3%)
Left/Right Split - PP: 70.1% / 29.9%
                   XS: 41.9% / 58.1%

(Assuming PP1 & PP3 = Left Foot, PP2 = Right Foot)
```

Summary

```
Load Distribution: How similar are left/right splits?

        Static:  PP = 53.4% L / 46.6% R   |   XS = 53.1% L / 46.9% R   |   Diff = 0.3 pp
  Weight Shift:  PP = 47.4% L / 52.6% R   |   XS = 50.1% L / 49.9% R   |   Diff = 2.7 pp
       Walking:  PP = 70.1% L / 29.9% R   |   XS = 41.9% L / 58.1% R   |   Diff = 28.2 pp

Average distribution difference across all trials: 10.4 percentage points
(Walking trial assumes PP1 & PP3 = Left, PP2 = Right)


Force Magnitude: How much higher does the Pressure Plate read?

        Static:  PP = 557.7 N,  XS = 457.6 N   ->   PP reads 100.1 N higher (18.0%)
  Weight Shift:  PP = 494.2 N,  XS = 415.3 N   ->   PP reads 78.9 N higher (16.0%)
       Walking:  PP = 276.5 N,  XS = 421.1 N   ->   PP reads 144.6 N lower (52.3%)

For static & weight shift trials, PP reads ~17% higher than XS on average
```